

Design Document

MVC:

- Patient, BillingEntry, and StaffUser are the Models
- The GUI (LoginFrame, MainWindow, NurseBillingDialog, DoctorBillingDialog) are the View
- The ClientHandler is the controller

```
private String handleCreatePatient(String[] parts) {  
    if (loggedInUser.isDoctor()) return err("Doctors cannot create patients");  
    if (parts.length < 2) return err("CREATE_PATIENT requires patient data");  
    Patient template = Patient.fromFileLine(parts[1]); // Factory method  
    if (template == null) return err("Malformed patient data");  
    int newId = model.createPatient(template);  
    return ok(String.valueOf(newId));  
}
```

Singleton:

- Used in HospitalServer
- Only one instance of the server exists

```
public static synchronized HospitalServer getInstance(int port) {  
    if (instance == null) {  
        instance = new HospitalServer(port);  
    }  
    return instance;  
}
```

Factory:

- Used in Patient
- Factory to create Patient Objects

```
public static Patient createNew(String firstName, String lastName) {  
    return new Patient(0, firstName, lastName, 0, "", 0.0, "", "", 1, "", false);  
}
```